Hydrothermal synthesis of MoS2-SiO2 core-shell particles

"What hydrothermal conditions favor MoS2 nanosheet growth on SiO2 cores?" *Generated by LeapSpace*

MoS_2 nanosheet growth on **SiO_2 cores** is favored by **hydrothermal temperatures around 180–200 °C, sufficient reaction time (about 18–24 h), and a precursor ratio that avoids excess Mo/S**, because these conditions promote **uniform nucleation, thin nanosheets, and good surface coverage** rather than agglomerated multilayers [11] [67] [68] [69] [70].

Conditions that favor nanosheet growth on SiO_2 cores

Hydrothermal factor	Favorable condition	Effect on MoS_2 growth on SiO_2
Temperature	**~180 °C** is repeatedly associated with good nanosheet formation; **200 °C** also gives well-formed MoS_2 in related hydrothermal syntheses [67] [68] [69] [70] [71]	Promotes **uniform nucleation** and crystalline nanosheet growth; too low limits growth, while too high tends to produce larger, stacked layers [68] [72]
Time	**18–24 h** is the best-supported window [67] [68] [69] [70] [71]	Gives enough time for nanosheets to develop; longer times can increase stacking and layer thickness [68] [73] [74]
Precursor amount / ratio	**Moderate Mo/S loading** is preferred; excess precursor causes overgrowth [68] [75] [76] [77]	Too little precursor leaves incomplete coverage, while too much drives **agglomeration** and thicker, less dispersed MoS_2 [68] [76]
pH / acidity	**Acidic conditions** are used in the clearest SiO_2-core example, and low pH is repeatedly linked to larger surface area and narrow pore distributions in MoS_2 hydrothermal growth [67] [75] [78]	Helps nucleation and self-assembly into nanosheet-based structures; low pH can also speed hydrolysis and weaken sheet integrity if pushed too far [78]
Core surface / template role	**SiO_2 as a hard template or reactive core** supports irregular nanosheet growth on the outer surface [68] [77]	The silica surface provides a scaffold for **anchoring and dispersing** MoS_2 nanosheets [68] [70]

What this means in practice

- A **thin, conformal, nanosheet-like MoS_2 shell** is more likely when the hydrothermal step is kept in the **180–200 °C** range and the reaction is held for **~20–24 h** [67] [68] [69] [70] [71].

- **Higher temperatures or longer times** tend to make the sheets **larger, thicker, and more stacked**, which reduces the benefit of nanosheet morphology [68] [73] [74].
- **Excess Mo and sulfur precursor** can push growth toward **agglomeration or sphere-like assemblies** rather than uniform nanosheet coverage [68] [76].
- The best-supported SiO_2-core route uses **hydrothermal growth followed by controlled shell formation**, not a highly concentrated or overextended reaction [67] [68].

Bottom line

For **MoS_2 nanosheet growth on SiO_2 cores**, the most favorable hydrothermal recipe is a **moderately acidic system at about 180–200 °C for roughly 18–24 h with controlled precursor loading**; this combination favors **uniform nucleation, thin nanosheets, and good shell coverage**, while harsher or more concentrated conditions promote **stacking and agglomeration** [67] [68] [68] [72] [75] [76].

Confidence level: **Medium** — the evidence consistently supports these hydrothermal trends, but the retrieved sources do not include a single dedicated optimization study specifically for MoS_2 grown on SiO_2 cores alone.

References

11. Kim, D.-H., Park, J.Y., Choi, H.S., (...), Baik, J.M. (2025) Advancing Humidity-Resistant Triboelectric Nanogenerators Through MoS_2-Encapsulated SiO_2 Nanoparticles for Self-Powered Gas Sensing Applications. *Advanced Energy Materials*. https://doi.org/10.1002/aenm.202405278

67. Guo, K., Shao, X., Chen, M., (...), Wang, L. (2026) Temperature-dependent tribological mechanisms of MoS2@SiO2 core-shell lubricants from 200 to 800 ℃: Experiments and molecular dynamics. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2026.112693

68. Jiang, X., Niu, H., Li, J., (...), Fan, B. (2023) Construction of core-shell structured SiO2@MoS2 nanospheres for broadband electromagnetic wave absorption. *Applied Surface Science*. https://doi.org/10.1016/j.apsusc.2023.157355

69. Li, Chia-Jui, Tseng, Chuan-Ming, Lai, Sz-Nian, (...), Hung, Wei-Hsuan (2017) *Nanoscale Research Letters*. https://doi.org/10.1186/s11671-017-2327-y

70. Jing, Yuting, Wang, Ruijing, Wang, Qiang, Wang, Xuefeng (2021) *Molecules*. https://doi.org/10.3390/molecules26247503

71. Manikandan, R., Raina, G. (2022) Hydrothermally synthesized 2H-MoS2 under optimized conditions - A structure and morphology analysis. *Physica Scripta*. https://doi.org/10.1088/1402-4896/ac9d6f

72. Habibi Jetani, G., Rahmani, M.B. (2022) Exploring the effect of hydrothermal precursor pH on the photosensitivity of 1T/2H–MoS2 nanosheets. *Optical Materials*. https://doi.org/10.1016/j.optmat.2022.111974

73. Zhang, D., Liang, S., Chai, J., (...), Cao, M. (2019) Highly effective shielding of electromagnetic waves in MoS2 nanosheets synthesized by a hydrothermal method. *Journal of Physics and Chemistry of Solids*. https://doi.org/10.1016/j.jpcs.2019.05.041

74. Huang, Y., Wang, S., Miao, X., (...), Yuan, S. (2025) Synergistic electrocatalytic reduction of Cr(VI) and oxidation of As(III) via an asymmetric electrode system for efficient wastewater treatment. *Journal of Environmental Management*. https://doi.org/10.1016/j.jenvman.2025.127718

75. Li, R., Liang, S., Aihemaiti, A., (...), Zhang, Z. (2023) Effectively enhanced piezocatalytic activity in flower-like 2H-MoS2 with tunable S vacancy towards organic pollutant degradation. *Applied Surface Science*. https://doi.org/10.1016/j.apsusc.2023.157461

76. Dong, T., Zhang, X., Wang, P., (...), Yang, P. (2020) Formation of Ni-doped MoS2 nanosheets on N-doped carbon nanotubes towards superior hydrogen evolution. *Electrochimica Acta*. https://doi.org/10.1016/j.electacta.2020.135885

77. Zhang, Z., Wang, P., Wang, F., (...), Yun, Z. (2018) Controlling dispersion and morphology of MoS2 nanospheres by hydrothermal method using SiO2 as template. *Chinese Journal of Chemical Engineering*. https://doi.org/10.1016/j.cjche.2017.12.016

78. Wang, W., Li, L., Wu, K., (...), Yang, Y. (2015) Hydrothermal synthesis of bimodal mesoporous MoS2 nanosheets and their hydrodeoxygenation properties. *RSC Advances*. https://doi.org/10.1039/c5ra09690a